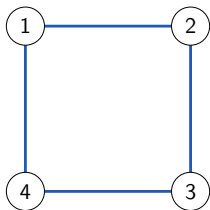


# Permutation Groups and Burnside's Lemma

Section 2.7.1–2.7.2

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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# Today's lecture

**1. Permutation Groups**

**2. Group Actions and Orbits**

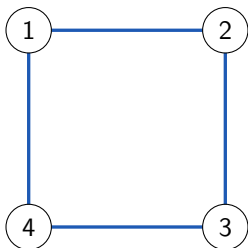
**3. Burnside's Lemma**

**4. Counting Up to Symmetry**

## Section 2.7.1: Permutation groups

### Question

How many different colorings of a square are there when rotations or reflections may make two colorings equivalent?



# Permutations as functions

## Permutation

A permutation of a finite set  $X$  is a bijection from  $X$  to itself. The permutations of  $[n]$  form the symmetric group  $S_n$ .

A permutation can be written in two-row notation or as a product of disjoint cycles.

# Cycle notation

## Example

The permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$$

is  $(1\ 3\ 2)(4\ 5)$ .

A cycle  $(a_1\ a_2\ \cdots\ a_k)$  sends  $a_i$  to  $a_{i+1}$  and  $a_k$  to  $a_1$ . Fixed points may be written as one-cycles or omitted.

# Composing permutations

## Convention

In a product  $\sigma\tau$ , apply  $\tau$  first and then  $\sigma$ .

## Example

Let  $\sigma = (1\ 2\ 3)$  and  $\tau = (1\ 2)$ . Then

$$\sigma\tau = (1\ 3), \quad \tau\sigma = (2\ 3).$$

In general, permutation multiplication is not commutative.

## Group

A set  $G$  with a multiplication is a group if multiplication is associative, there is an identity element  $e$ , and each element has an inverse. The product of elements of  $G$  must remain in  $G$ .

Permutations form groups under composition. The identity fixes every point, and the inverse reverses the bijection.

# Permutation groups

## Permutation group

A subgroup of  $S_n$  is a collection of permutations containing the identity and closed under composition and inverses.

Geometric symmetries act by permuting vertices, edges, faces, or other positions.

# Rotations of a square

Let  $r = (1\ 2\ 3\ 4)$ . The rotation group is

$$C_4 = \{e, r, r^2, r^3\}, \quad r^4 = e.$$

The rotations are by  $0^\circ, 90^\circ, 180^\circ, 270^\circ$  in the direction of the vertex labeling.

# All symmetries of a square

## Dihedral group

The group  $D_4$  has eight elements: four rotations and four reflections. If  $s$  is a reflection, then

$$D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\},$$

with  $r^4 = s^2 = e$  and  $srs = r^{-1}$ .

The allowed symmetry group is part of the counting problem.

## Action

An action of a group  $G$  on a set  $X$  assigns an element  $g \cdot x \in X$  to every  $g \in G, x \in X$ , with

$$e \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x).$$

For us,  $X$  will often be a set of colorings and  $G$  a geometric symmetry group.

# Acting on colorings

A coloring of a position set  $V$  with  $k$  colors is a function  $f : V \rightarrow [k]$ .

A permutation  $g$  of  $V$  acts by moving colors with their positions:

$$(g \cdot f)(v) = f(g^{-1}v).$$

This convention ensures  $(gh) \cdot f = g \cdot (h \cdot f)$ .

## Orbit

The orbit of  $x \in X$  is

$$\text{Orb}(x) = \{g \cdot x : g \in G\}.$$

Two objects are equivalent under the allowed symmetries exactly when they lie in the same orbit.

Orbits partition  $X$  into disjoint equivalence classes.

## Stabilizer

The stabilizer of  $x$  is

$$G_x = \{g \in G : g \cdot x = x\}.$$

It is a subgroup of  $G$ . Its elements are the symmetries that preserve the particular object  $x$ .

# The orbit–stabilizer theorem

## Theorem

For a finite group acting on a set,

$$|\text{Orb}(x)| = \frac{|G|}{|G_x|}.$$

Objects with more symmetry have smaller orbits.

# Proof of orbit–stabilizer

## Proof

Map  $G$  onto  $\text{Orb}(x)$  by  $g \mapsto g \cdot x$ .

If  $g \cdot x = h \cdot x$ , then  $h^{-1}g \in G_x$ , so the elements sending  $x$  to  $h \cdot x$  are precisely  $hG_x$ . Each fiber therefore has  $|G_x|$  elements.

The  $|G|$  elements of the group split into  $|\text{Orb}(x)|$  equal fibers.

## Why dividing by the group size fails

### Example

Color the square's vertices red or blue and allow rotations. An all-red coloring has orbit size 1. A coloring with just one red vertex has orbit size 4. An alternating coloring has orbit size 2.

Since orbit sizes vary, dividing the  $2^4$  labeled colorings by 4 does not count the orbits.

## Section 2.7.2: Burnside's lemma

### Fixed objects

For  $g \in G$ , write

$$\text{Fix}(g) = \{x \in X : g \cdot x = x\}.$$

### Burnside's lemma

The number of orbits of a finite group  $G$  on a finite set  $X$  is

$$\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

## Proof: count fixed pairs

### Proof

Count pairs  $(g, x) \in G \times X$  for which  $g \cdot x = x$ . Counting first by  $g$  gives

$$\sum_{g \in G} |\text{Fix}(g)|.$$

Counting first by  $x$  gives

$$\sum_{x \in X} |G_x|.$$

## Proof: group the objects by orbit

### Proof

Within an orbit  $O$ , orbit-stabilizer gives  $|G_x| = |G|/|O|$  for every  $x \in O$ . Therefore

$$\sum_{x \in O} |G_x| = |O| \frac{|G|}{|O|} = |G|.$$

Each orbit contributes exactly  $|G|$  fixed pairs. Divide the total by  $|G|$  to obtain the number of orbits.

## Fixed colorings and cycles

### Key observation

If a permutation of the positions has  $c$  cycles, including fixed points, then it fixes exactly  $k^c$  colorings with  $k$  available colors.

### Proof

A fixed coloring must be constant on each cycle. Choose its color independently for each of the  $c$  cycles.

## Square colorings up to rotation

| Rotation              | Cycles on vertices | Fixed colorings |
|-----------------------|--------------------|-----------------|
| Identity              | 4                  | $k^4$           |
| $90^\circ, 270^\circ$ | 1 each             | $k$ each        |
| $180^\circ$           | 2                  | $k^2$           |

$$\text{Number of orbits} = \frac{k^4 + k^2 + 2k}{4}.$$

## Two and three colors

With two available colors, rotation classes number

$$\frac{2^4 + 2^2 + 2 \cdot 2}{4} = 6.$$

With three available colors, they number

$$\frac{3^4 + 3^2 + 2 \cdot 3}{4} = 24.$$

The colors are labeled; a symmetry moves positions, not color names.

## Reflections of the square

| Reflection axis                 | Number | Cycles on vertices |
|---------------------------------|--------|--------------------|
| Through opposite vertices       | 2      | 3                  |
| Through opposite edge midpoints | 2      | 2                  |

The first type fixes two vertices and swaps the other two. The second type swaps two pairs.

# Square colorings up to all symmetries

## Formula

Under  $D_4$ , the number of vertex-coloring orbits is

$$\frac{k^4 + 2k^3 + 3k^2 + 2k}{8}.$$

For  $k = 3$ , this is 21. Allowing reflections merges some of the 24 rotation classes.

## A prescribed number of red vertices

### Question

How many square colorings with exactly two red and two blue vertices are there, up to all symmetries?

We must count fixed colorings satisfying the color inventory. The unrestricted count  $2^c$  is no longer appropriate.

## Exactly two red vertices: fixed counts

| Symmetry               | Number | Fixed colorings each |
|------------------------|--------|----------------------|
| Identity               | 1      | 6                    |
| Quarter-turn           | 2      | 0                    |
| Half-turn              | 1      | 2                    |
| Vertex-axis reflection | 2      | 2                    |
| Edge-axis reflection   | 2      | 2                    |

Burnside gives  $(6 + 2 + 4 + 4)/8 = 2$  orbits: red vertices adjacent or opposite.

# Necklaces with $n$ positions

## Rotation equivalence

A necklace here is a coloring of  $n$  cyclic positions, with rotations identified but reflections kept distinct.

Rotation by  $j$  positions has  $\gcd(n, j)$  cycles, taking  $\gcd(n, 0) = n$ .

$$\text{Number of necklaces} = \frac{1}{n} \sum_{j=0}^{n-1} k^{\gcd(n, j)}.$$

# Why the gcd counts cycles

## Proof

Under rotation by  $j$ , a position returns after the least positive number  $\ell$  such that  $\ell j \equiv 0 \pmod{n}$ . This is  $\ell = n / \gcd(n, j)$ .

All cycles have this length, so their number is  $n/\ell = \gcd(n, j)$ .

# Prime-length necklaces

## Formula

For prime  $p$ , the number of  $k$ -color necklaces is

$$\frac{k^p + (p - 1)k}{p}.$$

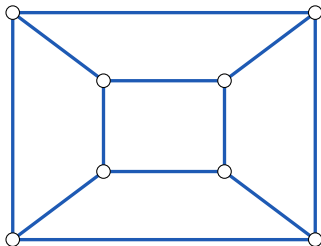
The identity has  $p$  cycles, and every nonidentity rotation has one cycle.

Since the count is integral,  $p$  divides  $k^p - k$ .

# Coloring the faces of a cube

## Question

How many face colorings of a cube are there up to rotation, using  $k$  available colors?



# The cube rotation group

There are 24 rotations: choose the image of one face in 6 ways, then the orientation around its normal in 4 ways.

| Rotation type           | Number | Face-cycle lengths |
|-------------------------|--------|--------------------|
| Identity                | 1      | 1, 1, 1, 1, 1, 1   |
| Face axis, quarter-turn | 6      | 1, 1, 4            |
| Face axis, half-turn    | 3      | 1, 1, 2, 2         |
| Vertex axis, third-turn | 8      | 3, 3               |
| Edge axis, half-turn    | 6      | 2, 2, 2            |

## Applying Burnside to cube faces

### Formula

The number of face-coloring orbits under rotations is

$$\frac{k^6 + 3k^4 + 12k^3 + 8k^2}{24}.$$

With two colors, the answer is

$$\frac{64 + 48 + 96 + 32}{24} = 10.$$

# A checklist for orbit counting

- ① Specify the objects and the allowed symmetry group.
- ② Classify the group elements by their action on positions.
- ③ Count the objects fixed by each element.
- ④ Average the fixed counts using Burnside's lemma.

If color counts or other restrictions are imposed, include them in step 3.

## Question

How many colorings of the vertices of an equilateral triangle with  $k$  colors are there up to rotation? What changes if reflections are also allowed?

## Practice: solution

For rotations, the identity fixes  $k^3$  colorings and the two nonidentity rotations each fix  $k$ :

$$\frac{k^3 + 2k}{3}.$$

For all six symmetries, the three reflections each have two vertex cycles and fix  $k^2$  colorings:

$$\frac{k^3 + 3k^2 + 2k}{6} = \binom{k+2}{3}.$$

- Group actions describe equivalence under symmetry.
- Orbit–stabilizer relates an object’s symmetries to its orbit size.
- Burnside’s lemma counts orbits by averaging fixed objects.
- For unrestricted colorings, a permutation with  $c$  position cycles fixes  $k^c$  colorings.

# Partitions and Special Numbers